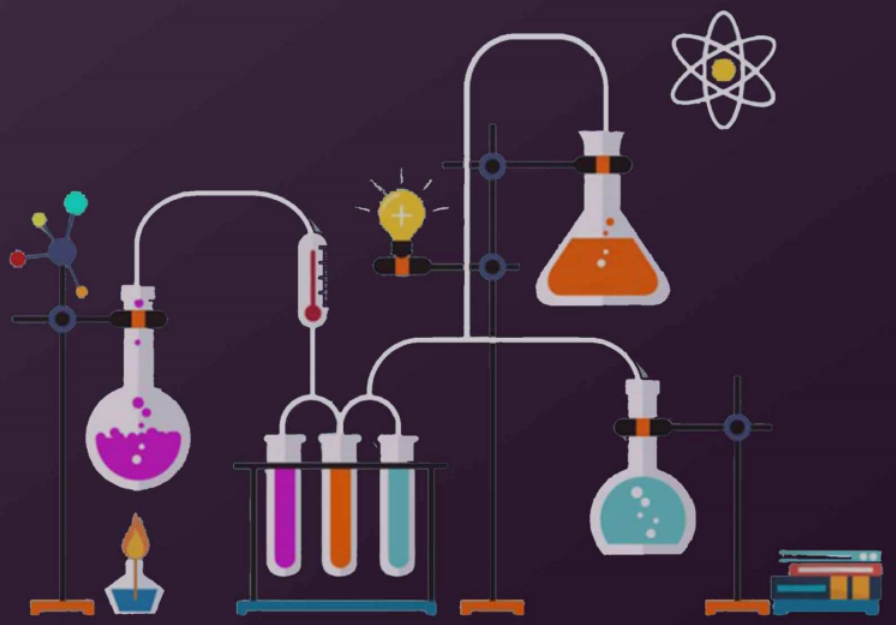
CHEMISTRY

Chemical Equilibrium



SHEET

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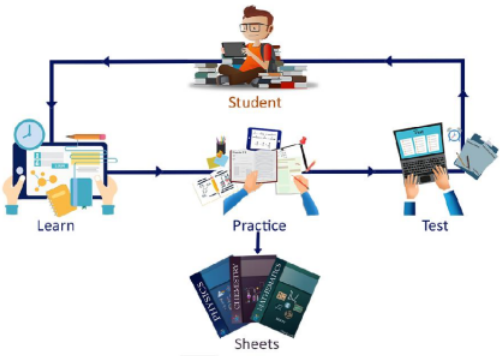
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Chemical Equilibrium	1 - 3			
2	Equilibrium Constant and Characteristics of Equilibrium	4 - 14	1 - 4, 10,11, 14	1 - 8	2,4
3	Active Mass and Types of Equilibrium Constants	15 - 25	5	9,10	
4	Problems on Equilibrium Constants 1	26 - 37	12	11-27	1
5	Problems on Equilibrium Constants 2	38 - 45	6	28-35	6,7,12,13
6	Factors Affecting Equilibrium Constant	46 - 52	7 - 8	36-41	9
7	LeChatelier's Principle	53 - 57	13	42-44	
8	Application of LeChatelier's Principle			45-49	5,8,10
9	ACC		9,15	50-57	3,11

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Chemical Equilibrium

Q1. In which of the following does the reaction go nearest to completion ?

(A) $K_{eq} = 10^{6}$ (B) $K_{eq} = 10^{3}$

(C) $K_{eq} = 10^{-6}$ (D) $K_{eq} = 10^{-12}$

Q2. A chemical reaction is at equilibrium when :-

(A) Reactants are completely transformed into products

(B) The rates of forward and backward reactions are equal

(C) Formation of products is minimised

(D) Equal amounts of reactants and products are present

Q3. In which of the following reaction equilibrium can not be established ?

(A) $NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

(B) $HCl + NaOH \longrightarrow NaCl + H_2O$

(C) $H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

(D) All of these

Equilibrium Constant and Characteristics of Equilibrium

Q4. For the reaction $A(g) + 3B(g) \square 2C(g)$ at 27°C, 2 moles of A, 4 moles of B and 6 moles of C are present in 2 litre vessel. If K_C for the reaction is 1.2, the reaction will proceed in :-

(A) Forward direction

(B) Backward direction

(C) Neither direction

(D) None of these

Q5. The equilibrium

$2SO_2(g) + O_2(g) \square 2SO_3(g)$ shifts forward if :-

(A) A catalyst is used.

(B) An adsorbent is used to remove SO_3 as soon as it is formed.

(C) Small amounts of reactants are removed.

(D) None of these

Q6. Consider following reactions in equilibrium with equilibrium concentration 0.1 M of every species:-

(I) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(II) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(III) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Extent of reactions taking place is :-

(A) I > II > III (B) I < II < III

(C) II < III < I (D) III < I < II

Q7. The reaction $PCl_{5(s)} \longrightarrow PCl_{3(g)} + Cl_{2(g)}$ is in equilibrium. If equilibrium concentration of $PCl_{3(g)}$ is doubled, then concentration of $Cl_{2(g)}$ would become

(A) 1/4 of its intial value

(B) 1/2 of its intial value

(C) Two times of its initial value

(D) Four times of its initial value

Q8. Consider the reactions :

(i) $2CO(g) + 2H_2O(g) \rightarrow 2CO_2(g) + 2H_2(g)$; Eqm. constant = K_1

(ii) $CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$; Eqm. Constant = K_2

(iii) $CH_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4H_2(g)$; Eqm. Constant = K_3

Which of the following relation is correct ?

(A) $K_3 = \frac{K_1}{K_2}$ (B) $K_3 = \frac{K_1^2}{K_2^2}$

(C) $K_3 = K_1K_2$ (D) $K_3 = \sqrt{K_1} \cdot K_2$

Q9. For the system, $3A + 2B \longrightarrow C$, the expression for equilibrium constant is :-

(A) $\frac{[A]^3[B]^2}{[C]}$ (B) $\frac{[C]}{[A]^3[B]^2}$

(C) $\frac{[A]^2[B]^3}{[C]}$ (D) $\frac{[C]}{[A][B]}$

Q10. For the reaction $2A(g) \longrightarrow B(g) + 3C(g)$, at a given temperature, $K_c = 16$. What must be the volume of the flask, if a mixture of 2 mole each of A, B and C exist in equilibrium ?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) 1 (D) None of these

Q11. For reaction $HI \rightarrow \frac{1}{2} H_2 + \frac{1}{2} I_2$ value of K_C is 1/8 then value of K_C for $H_2 + I_2 \rightarrow 2HI$.

(A) $\frac{1}{64}$ (B) 64

(C) $\frac{1}{8}$ (D) 8

Q12. The equilibrium constant (K_C) for the reaction $N_2(g) + O_2(g) \longrightarrow 2NO(g)$ at temperature T is 4×10^{-4}. The value of K_C for the reaction.

$NO(g) \longrightarrow \frac{1}{2}N_2(g) + \frac{1}{2}O_2(g)$ at the same temperature is :-

(A) 50.0 (B) 0.02

(C) 2.5×10^2 (4) 4×10^{-4}

Q13. Using molar concentrations what is the unit of K_C for the reaction :

$CH_3OH(g) \square CO(g) + 2H_2(g)$:-

(A) M^{-2} (B) M^2

(C) M^{-1} (D) M

Q14. Write the relation between K_p & K_c for the reaction:

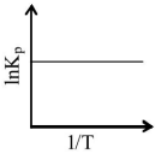
$C(s) + O_2(g) \square CO_2(g)$

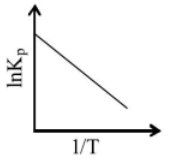
(A) $K_p = K_c(RT)^{-1}$ (B) $K_p = K_c(RT)^1$

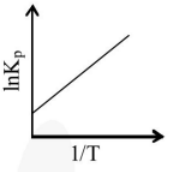
(C) $K_p.K_c = 1$ (D) $K_p = K_c$

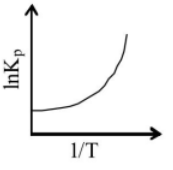
Active Mass and Type of Equilibrium Constants

Q15. An endothermic reaction is represented by the graph -

(A) 

(B) 

(C) 

(D) 

Q16. 10 moles of SO_3 gas is taken in 1 ℓt. closed rigid container and allowed to attain equilibrium at 27°C as :

$2SO_{3(g)} \longrightarrow 2SO_{2(g)} + O_{2(g)}$

2 moles of $O_{2(g)}$ is formed at equilibrium. If 10 moles of $SO_{2(g)}$ & 5 moles of $O_{2(g)}$ is taken in the same container at same temperature, then number of moles of $SO_{3(g)}$ formed at equilibrium is :

(A) 4 mole (B) 8 mole

(C) 2 mole (D) 6 mole

Q17. In a reaction $A + 2B \rightleftharpoons 2C$. If 2 mole of A, 3 mole of B and 2 mole of C are placed in a 2 L flask and the equilibrium 4 mole of C were found then find at the value of K_C-

(A) 16 (B) 32 (C) 8 (D) 24

Q18. For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$;

$K_C = \frac{\alpha^2}{(1-\alpha)V}$, keeping temperature constant.

(A) K_C will increase on increasing the volume

(B) K_C will decrease on increasing the volume

(C) K_C & α will not change on changing the volume

(D) K_C will not change but α will change on changing the volume

Q19. The equilibrium constant (K_C) for the reaction $2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g)$ is 4×10^{-34} at 25°C what is the equilibrium constant for reaction :

$$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightleftharpoons HCl(g)$$

(A) 2×10^{-17} (B) 2.5×10^{33}
(C) 5×10^{6} (D) 2.5×10^{21}

Q20. For the reaction

$CuSO_4.5H_2O(s) \rightleftharpoons CuSO_4.3H_2O(s) + 2H_2O(g)$

Which one is correct representation -

(A) $K_p = P^2_{H_2O}$ (B) $K_c = [H_2O]^2$
(C) $K_p = K_c(RT)^2$ (D) All

Q21. For the reaction

$$CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g) \quad \frac{K_P}{K_C} \text{ is :-}$$

(A) RT (B) $(RT)^{-1}$
(C) $(RT)^{-1/2}$ (D) $(RT)^{1/2}$

Q22. $A + B \rightleftharpoons C + D$. If finally the concentration of A and B are both equal but at equilibrium concentration of D will be twice of that of A then what will be the equilibrium constant of reaction:-

(A) 4 / 9 (B) 9 / 4
(C) 1 / 9 (D) 4

Q23. At 273 K the value of K_p for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 0.2. What is the value of K_c at 273 K -

(A) 0.4 (B) 0.1
(C) 0.2 (D) 0.6

Q24. For which of following equilibrium the value of K_p is less than K_C :-

(A) $H_2 + I_2 \rightleftharpoons 2HI$

(B) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(C) $N_2 + O_2 \rightleftharpoons 2NO$

(D) $CO + H_2O \rightleftharpoons CO_2 + H_2O$

Q25. When 1 mole of N_2 and 3 mole of H_2 was heated in a vessel at 773 K and a total pressure is 3.4 atm. 30% N_2 is converted into NH_3. At equilibrium calculate the K_P for the reaction :

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

(A) $\frac{(0.6)^2}{(0.7)(2.1)^3}$ (B) $\frac{(0.7)^2}{(0.6)(2.1)^3}$

(C) $\frac{(2.1)^2}{(0.6)(0.7)^3}$ (D) $\frac{(0.3)^2}{(0.6)(0.7)^3}$

Problems on Equilibrium Constants 1

Q26. $xA_{(s)} \rightleftharpoons yB_{(g)} + zC_{(g)}$

If $\frac{k_c}{k_p} = (RT)^{-2}$, then which is correct -

(A) $y + z - x = -2$ (B) $y + z - x = 2$
(C) $y + z = -2$ (D) $y + z = 2$

Q27. For the reversible reaction in equilibrium

$$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \underset{k_2}{\overset{K}{\rightleftharpoons}} NO(g)$$

$C_0 = Ce^{-2.1 \times 10^{-3} t}$ for the forward reaction and $C'_0 = C'e^{-4.2 \times 10^{-4} t}$ for the backward reaction, hence K_c for the above equilibrium is -

(A) 5.0 (B) 2.0
(C) 0.5 (D) 2

Q28. 1.1 mol of A & 2.2 mol of B were taken initially in a 1L flask $A + 2B \rightleftharpoons 2C + D$. At equilibrium 0.2 mol of C is formed. The equilibrium constant of the reaction is :

(A) 0.001 (B) 0.002
(C) 0.003 (D) 0.004

Q29. For the reaction $2HI \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociation α of HI(g) is related to equilibrium constant K_P by the expression :-

(A) $\frac{1+2\sqrt{K_P}}{2}$ (B) $\sqrt{\frac{1+2K_P}{2}}$

(C) $\sqrt{\frac{2K_P}{1+2K_P}}$ (D) $\frac{2\sqrt{K_P}}{1+2\sqrt{K_P}}$

Q30. For the reaction (A) and (B) :

$A \rightleftharpoons B + C$ ....(A)

$D \rightleftharpoons 2E$ ....(B)

Given $K_{P_1} : K_{P_2} :: 9 : 1$

If the degree dissociation of A and D be same then the total pressure at equilibria (A) and (B) are in the ratio (Assume reaction are started with equal number of moles of A and D).

(A) 3 : 1 (B) 36 : 1

(C) 1 : 1 (D) 0.5 : 1

Q31. The value of K_p for the following reaction $2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$, is 1.2×10^{-2} at 106.5°C. The value of K_c for this reaction is :-

(A) $= 1.2 \times 10^{-2}$ (B) $< 1.2 \times 10^{-2}$

(C) $> 1.2 \times 10^{-2}$ (D) None of these

Q32. For the reaction :

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \quad ; K_P$$

The degree of dissociation (α) of NH_3 is related to total equilibrium pressure (P^o) as :

(A) $\alpha = \left(1+\frac{3\sqrt{3}\,P^o}{4K_P}\right)^{-\frac{1}{2}}$

(B) $\alpha = \left(1+\frac{3\sqrt{3}\,P^o}{4K_P}\right)^{\frac{1}{2}}$

(C) $\alpha = \left(1+\frac{3P_0}{4K_p}\right)^{\frac{1}{2}}$

(D) $\alpha = \left(1+\frac{3P_0}{4K_p}\right)^{-\frac{1}{2}}$

Q33. PCl_5 is 40% dissociated when pressure is 2 atm. It will be 80% dissociated when pressure is (approx) :

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) 0.31 atm (B) 0.51 atm

(C) 0.61 atm (D) 0.21 atm

Q34. The vapour denisty of a mixture containing NO_2 and N_2O_4 is 27.6. The mole fraction of N_2O_4 in the mixture is :-

(A) 0.1 (B) 0.2 (C) 0.5 (D) 0.8

Q35. For the reaction

$3A(g) + 2B(g) \rightleftharpoons 4C(g) + D(g) + 2E(g)$.

The unit of K_p is -

(A) $(atm)^2$ (B) $(atm)^{-2}$

(C) $(atm)^5$ (D) $(atm)^{-5}$

Q36. K_p/K_C for the reaction :

$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$ is :-

(A) $(RT)^{-1}$ (B) RT

(C) $\frac{1}{\sqrt{RT}}$ (D) $(RT)^{\frac{1}{2}}$

Q37. For the equilibrium of the reaction,

$HgO(s) \rightleftharpoons Hg(g) + \frac{1}{2}O_2(g)$, k_p for the reaction at total pressure of P is :

(A) $K_P = \frac{2}{3^{3/2}}p^{3/2}$ (B) $K_P = \frac{2}{3^{1/2}}p^{1/2}$

(C) $K_P = \frac{2}{3^{2/3}}p^{3/2}$ (D) $K_P = \frac{1}{3^{2/3}}p$

Problems on Equilibrium Constants 2

Q38. For the reaction, $N_2 + O_2 \rightleftharpoons 2NO$ equilibrium constant $K_c = 2$. Degree of dissociation of N_2 and O_2 are : (Both have same initial moles)

(A) $\frac{1}{1+\sqrt{2}}, \frac{1}{1-\sqrt{2}}$ (B) $\frac{1}{1-\sqrt{2}}, \frac{1}{1+\sqrt{2}}$

(C) Both are $\frac{1}{1+\sqrt{2}}$ (D) Both are $\frac{1}{1-\sqrt{2}}$

Q39. A mixture of 0.3 mole of H_2 and 0.3 mole of I_2 is allowed to react in a 10 L evacuated flask at 500°C.

The reaction is $H_2(g) + I_2 \rightleftharpoons 2HI(g)$, the K is found to be 64. The amount of unreacted I_2 at equilibrium is :-

(A) 0.03 mole (B) 0.06 mole

(C) 0.09 mole (D) 3.6 moles

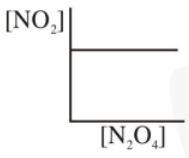
Q40. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

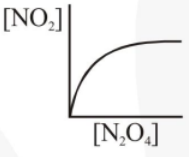
For the reaction initially the mole ratio was 1 : 3 of N_2 : H_2. At equilibrium 50% of each has reacted. If the equilibrium pressure is P, the partial pressure of NH_3 at equilibrium is :-

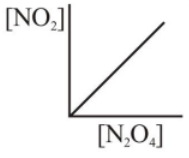
(A) $\frac{P}{3}$ (B) $\frac{P}{4}$ (C) $\frac{P}{6}$ (D) $\frac{P}{8}$

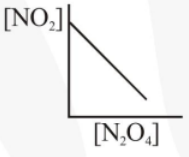
Q41. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

The concentration of $N_2O_4(g)$ and of $NO_2(g)$ for which the following reactions will be at equilibrium will lie on following graph :-

(A) 

(B) 

(C) 

(D) 

Q42. The equilibrium constant K_p for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is 1.6 at 200°C. The pressure at which PCl_5 will be 50% dissociated at 200°C :-

(A) 3.2 atm (B) 4.8 atm
(C) 2.4 atm (D) 6.4 atm

Q43. For the reaction equilibrium ;

$2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$,

if $P_{Br_2} = \frac{P}{9}$ at equilibrium and P is total pressure, the ratio $\frac{K_p}{P}$ is equal to :-

(A) $\frac{1}{9}$ (B) $\frac{1}{81}$ (C) $\frac{1}{27}$ (D) $\frac{1}{3}$

Q44. Some solid NH_4HS is placed in a flask containing NH_3, whose pressure is 0.5 atm what would be pressures of NH_3 when equilibrium is reached :

$NH_4HS_{(g)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$, $K_p = 0.11$

(A) 6.65 atm (B) 0.665 atm
(C) 0.0665 atm (D) 66.5 atm

Q45. For the reaction

$C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

$K_p = 5\times10^{-2}$ atm. Calculate the mole per cent of $C_2H_6(g)$ at equilibrium if pure C_2H_6 at 1 atm is passed over a suitable catalyst at 900 K :-

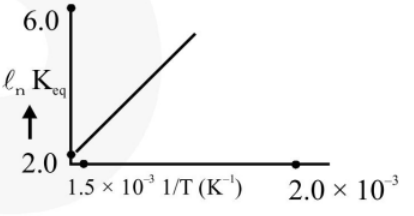
(A) 20 (B) 33.33
(C) 66.66 (D) 25.1

Factors Affecting Equilibrium Constant

Q46. Which is in favour of forward reaction for reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ at constant temperature :-

(A) Addition of inert gas at constant volume
(B) Addition of $PCl_3(g)$ at constant volume
(C) Addition of $PCl_5(g)$ at constant volume
(D) Addition of $Cl_2(g)$ at constant volume

Q47. A schematic plot of $\ell n\, K_{eq}$ verses inverse of temperature for a reaction is shown below. The reaction must be :



(A) endothermic
(B) exothermic
(C) highly spontaneous at ordinary temperature
(D) one with negligible enthalpy change

48. a moles of PCl_5, undergoes thermal dissociation as :

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the mole fraction of PCl_3 at equilibrium is 0.30 & the total pressure is 5 atm. Then partial pressure of Cl_2 at equilibrium is :

(A) 2.5 atm (B) 1.5 atm
(C) 0.5 atm (D) None

Q49. Which of the following statement is correct ?
(A) Catalyst affects the equilibrium position
(B) Chemical equilibrium is also known as dynamic equilibrium
(C) At equilibrium Q > Kc
(D) At equilibrium Q < Kc

Q50. The chemical reaction:

$BaO_{2(s)} \rightleftharpoons BaO_{(s)} + O_{2(g)}$, ΔH = +ve.
In equilibrium condition, pressure of O_2 depends upon :-
(A) Increase mass of BaO
(B) Increase mass of BaO_2
(C) Increase in temperature
(D) Increase mass of BaO_2 and BaO both

Q51. In the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)} -$ 180.7 kJ, on increasing the temperature, the production of NO.
(A) Increases
(B) Decreases
(C) Remains same
(D) Cannot be predicted

Q52. In a equilibrium reaction :

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
ΔH = – 3000 calories,
which factor favours dissociation of HI :-
(A) Low temp.
(B) High Pressure
(C) High temp.
(D) Low pressure.

Le Chatelier's Principle

Q53. In the equilibrium, $AB(g) \rightleftharpoons A(g) + B(g)$, if the equilibrium concentration of A is doubled, the equilibrium concentration of B would become :-
(A) half (B) twice
(C) $\frac{1}{4^{th}}$ (D) $\frac{1}{8^{th}}$

Q54. In an aqueous solution of volume 500 mL, when the reaction of

$2Ag^+ + Cu_{(s)} \rightleftharpoons Cu^{2+} + 2Ag_{(s)}$ reached equilibrium the $[Cu^{2+}]$ was xM. When 500 mL of water is further added, at the new equilibrium $[Cu^{2+}]$ will be :
(A) 2 xM
(B) xM
(C) Between xM and x/2M
(D) Less than x/2M

Q55. For which of the following reactions, the equilibrium position will not be affected by increase in pressure.

(I) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$
(II) $2NH_3(g) \rightleftharpoons H_2(g) + 3H_2(g)$
(III) $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$
(IV) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
(A) I and III (B) II and IV
(C) I and II (D) III and IV

Q56. For which of the following reactions is product formation favoured by low pressure and high temperature ?

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$; ΔH° = – 9.4 kJ
(B) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$; ΔH° = 172.5 kJ
(C) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH$; ΔH° = –21.7kJ
(D) $3O_2(g) \rightleftharpoons 2O_3(g)$; ΔH° = 285 kJ

Q57. Which of the following factors will be favourable for higher yields of NO in the reaction given below :
$N_2 + O_2 \rightleftharpoons 2NO(g)$ (ΔH = +ve)
(A) Low temperature, high pressure and high concentration of N_2 and O_2.
(B) High temperature and high concentration of N_2 and O_2. The reaction remains unaffected by pressure.
(C) High temperature, low pressure and low concentration of N_2 and O_2.
(D) Low temperature, low pressure and high concentrations of N_2 and O_2.

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Choose the incorrect statements, for reacton R $\longrightarrow$ P (irreversible reaction) :-

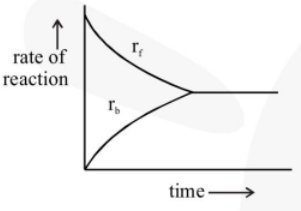
(A) Thermodynamically $\Delta rG^\circ >>> 0$ for above reaction

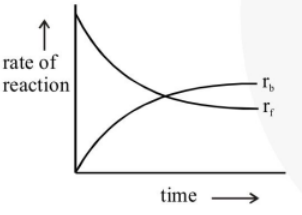
(B) Kinetically $K_f >>> K_b$ and $E_af >>> E_ab$

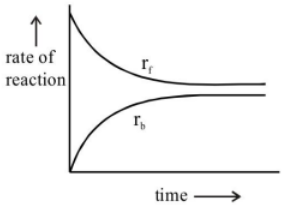
(C) Decomposition of solid $CaCO_3$ is example of irreversible reaction in open vessel $CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)\uparrow$

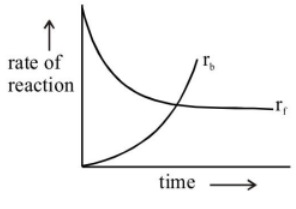
(D) Kinetically $K_f >> K_b$ and $E_{af} <<< E_{ab}$.

Q2. Which of the following represent the rates of forward reaction (r_f) and rates of backward reaction (r_b) at equilibrium ?

(A)


(B)


(C)


(D)


Q3. K_{eq} of a reaction $aA + bB \rightleftharpoons cC + dD$ can be expressed in the form of :-

(A) $K_{eq} = \dfrac{[cC][dD]}{[aA][bB]}$ (B) $K_{eq} = \dfrac{[C]^c[D]^d}{[A]^a[B]^b}$

(C) $K_{eq} = \dfrac{[A]^d[B]^b}{[C]^c[D]^d}$ (D) $\dfrac{1}{K_{eq}} = \dfrac{[A]^a[B]^b}{[C]^c[D]^d}$

Q4. For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

The value of equilibrium constant K_p doesn't changes with :-

(A) Total pressure

(B) Catalyst

(C) The amounts of H_2 and I_2 present

(D) Temperature

Q5. The following equilibrium are given :-

$N_2 + 3H_2 \rightleftharpoons 2NH_3$, K_1

$N_2 + O_2 \rightleftharpoons 2NO$, K_2

$H_2 + \dfrac{1}{2}O_2 \rightleftharpoons H_2O$, K_3

The equilibrium constant of the reaction

$2NH_3 + \dfrac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$

(A) $\dfrac{K_1}{K_2}$ (B) $\dfrac{K_1K_3}{K_2}$

(C) $\dfrac{K_2K_3^3}{K_1}$ (D) $K_1 > K_2 > K_3$

Q6. For given two equilibria attained in a container which are correct if degree of dissociation of A and A' are α and α' :-

$A(s) \rightleftharpoons 2B(g) + C(g)$ $K_{P_1} = 8\times10^{-2}$

$A'(s) \rightleftharpoons 2B(g) + D(g)$ $K_{P_2} = 2.0\times10^{-2}$

(A) $\dfrac{K_{P_1}}{K_{P_2}} = \left[\dfrac{(3\alpha'+2\alpha)}{(3\alpha+2\alpha')}\right]^3 \times \dfrac{\alpha}{\alpha'}$

(B) $P_B' = 2P_C' + 2P_D'$

(C) $\dfrac{P_C'}{P_B'} = 4$

(D) $\alpha > \alpha^1$

Q7. Chonsider two reactions X and Y

$X = A(g) \rightleftharpoons 2B(g), \Delta n_g > 0$

$Y = 2B(g) \rightleftharpoons A(g), \Delta n_g < 0$

When we increase pressure at equilibrium. Choose correct option(s) :-

(A) Reaction X moves forward

(B) Reaction X moves backward

(C) Reaction Y moves forward

(D) Reaction Y moves backward

Q8. For the given reaction, the correct statement(s) is (are) :

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)\ ; K_{eq} = K$

(A) Increase in $[CaCO_3]$ increase yield.

(B) K is independent of initial amount of $CaCO_3$.

(C) K is dependnt on pressure of CO_2 at given T.

(D) ΔH is independent of the catalyst.

Q9. On a phase diagram, the interaction of the solid, liquid and gaseous lines is known as the :-

(A) Intersection point

(B) Triple point

(C) Fluid point

(D) None of these

PARAGRAPH TYPE

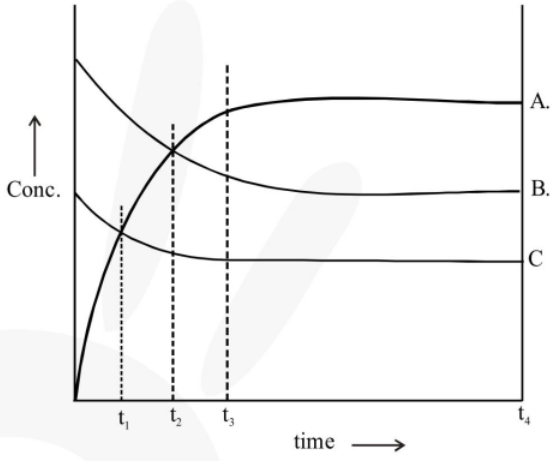
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.10 to 11

In a reversible chemical reaction, the rate of forward reaction decreases and that of backward reaction increases with the passage of time ; at equilibrium the rate of forward and backward reactions become same. Let us consider the formation of $SO_3(g)$ in the following reversible reaction :

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Following graphs are plotted for this reactions :



Q10. In the above graph, A B & C respectively are

(A) SO_3, SO_2 and O_2

(B) SO_3, O_2 and SO_2

(C) SO_2, O_2 and SO_3

(D) O_2, SO_2, and SO_3

Q11. In the above graph, the equilibrium state is attained at time :

(A) t_1 (B) t_2

(C) t_3 (D) t_4

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q12. Match list I with list II and select the correct answer:

List–I	**List–II**
Reaction	**Degree of dissociation 'α' (assume α <<1)**
(A) $2X(g) \rightleftharpoons 3Y(g) + Z(g)$	(P) $\alpha = \sqrt{\frac{K_C}{V}} \times 3$
(B) $4X(g) \rightleftharpoons Y(g) + Z(g)$	(Q) $\alpha = [2K_C V]^{1/3}$
(C) $2X(g) \rightleftharpoons Y(g) + 2Z(g)$	(R) $\alpha = \frac{4}{V}\sqrt{K_C}$
(D) $3X(g) \rightleftharpoons Y(g) + Z(g)$	(S) $\alpha = \left[\frac{16V^2K_C}{27}\right]^{1/4}$

(A) A→Q ; B→P ; C→R ; D→S
(B) A→R ; B→P ; C→S ; D→Q
(C) A→P ; B→R ; C→S ; D→Q
(D) A→Q ; B→R ; C→P ; D→S

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13.

Column–I	**Column–II**
(A) $A(g) + B(g) \rightleftharpoons AB(g)$	(P) Forward shift by addition of inert gas at constant pressure
(B) $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$	(Q) Backward shift by addition of inert gas at constant pressure
(C) $A(s) \rightleftharpoons B(g) + C(g)$	(R) Not affected by addition of inert gas at constant volume and also unaffected by increase in pressure
(D) $\frac{1}{2}A_2(g) + \frac{3}{2}B_2(g) \rightleftharpoons AB_3(g)$	(S) Not affected by addition of small amount of reactant

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q14. Consider following reaction :

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

[Reaction is carried out in close vessel]

How many of properties which become constant at equilibrium :-

(1) total moles (2) pressure

(3) colour (4) magnetism

(5) volume (6) $M_{avg.} = \sum X_i M_i$

(7) mole fraction

(8) density of mixture

Q15. Number of reaction in which Δn_g is greater than zero :-

(1) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(2) $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

(3) $H_2O(g) \rightleftharpoons H_2O(\ell)$

(4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3$

(5) $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

(6) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(7) $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

56 g of nitrogen and 8 g of hydrogen gas are heated in a closed vessel. At equilibrium, 34 g of ammonia are present. The equilibrium number of moles of nitrogen, hydrogen and ammonia are respectively :-

(A) 1, 2, 2 (B) 2, 2, 1

(C) 1, 1, 2 (D) 2, 1, 2

Q2. A chemical reaction A $\rightleftharpoons$ B is said to be in equilibrium when :

(A) Rate of forward reaction is equal to rate of backward reaction

(B) Conversion of A to B is only 50% complete

(C) Complete conservation of A to B has taken place

(D) Only 25% conversion of A to B has taken place

Q3. For which of the following reaction the relation $K_C = K_pRT$ is correct ?

(A) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(B) $PCl_3(g) + Cl_2(g) \rightarrow PCl_5(g)$

(C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(D) None of the above

Q4. The expression of K_C for the reaction,

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ is :

(A) $K_C = [CO] \times [O_2]^{1/2}$

(B) $K_C = \frac{[CO]}{1/2[O_2]}$

(C) $K_C = \frac{[CO]}{[O_2]^{1/2}}$

(D) $K_C = \frac{[CO]}{[C]+[O_2]^{1/2}}$

Q5. K_c for the reaction,

$2X(g) + Y(s) \rightleftharpoons P(g) + Q(g)$ is :-

(A) $K_c = \frac{[P][Q]}{[X]^2[Y]}$ (B) $K_c = \frac{[P]+[Q]}{[X]^2}$

(C) $K_c = \frac{[P][Q]}{[X]^2}$ (D) $K_c = \frac{[P][Q]}{[Y]}$

Q6. Which one is the correct representation for, $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$?

(A) $K_P = \dfrac{\left[P_{so_3}\right]^2}{\left[P_{so_2}\right]^2[P_{o_2}]}$

(B) $K_C = \dfrac{[SO_3]^2}{[SO_2]^2[O_2]}$

(C) $K_P = \dfrac{[n_{SO_3}]^2}{[n_{SO_2}]^2[n_{O_2}]} \times \left[\dfrac{p}{\text{Total mole}}\right]^{-1}$

(D) All of the above

Q7. The unit of K_C of a reaction is $mol^{-2}L^2$. For this reaction, the product concentration increases by ; if reaction is exothermic.

(A) Lowering the pressure

(B) Increase in pressure

(C) Lowering the temperature

(D) Both B & C

Q8. The correct relation between K_p & K_c for the reaction $a\,A + b\,B \rightleftharpoons b\,C + a\,D$ is :

(A) $K_p = K_c (RT)^{a+b}$ (B) $K_p = K_c$

(C) $K_p = K_c(RT)^1$ (D) $K_p = \dfrac{K_C}{(a+b)^2}$

Q9. Consider the following reactions in which all the reactants and products are in gaseous state :

$2PQ \rightleftharpoons P_2 + Q_2$; $K_1 = 2.5 \times 10^5$

$PQ + \frac{1}{2}R_2 \rightleftharpoons PQR$; $K_2 = 5 \times 10^{-3}$

The value of equilibrium constant for reaction :

$\frac{1}{2}P_2 + \frac{1}{2}Q_2 + \frac{1}{2}R_2 \rightleftharpoons PQR$ is :

(A) 2.5×10^{-3} (B) 2.5×10^3

(C) 1×10^{-5} (D) 5×10^3

Q10. The equilibrium constant for the reaction :

$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$; K_1

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$; K_2

Then relation between K_1 and K_2 is :

(A) $K_1 = K_2^2$ (B) $K_1 = \dfrac{1}{K_2^2}$

(C) $K_1 = \sqrt{K_2}$ (D) $K_1 = \dfrac{1}{\sqrt{K_2}}$

Q11. 5 moles each of hydrogen and iodine were heated in a sealed 10 litre vessel. At equilibrium, 2 moles of HI were found. The equilibrium constant for the reaction :

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is :

(A) 1 (B) 10

(C) 5 (D) 0.25

Q12. Determine the value of equilibrium constant (K_C) for the reaction

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

If 10 moles of A_2 ; 15 moles of B_2 and 5 moles of AB are placed in a 2 litre vessel and allowed to come to equilibrium. The final concentration of AB is 7.5 M :-

(A) 4.5 (B) 1.5

(C) 0.6 (D) None of these

Q13. Given $[CS_2]$= 0.120 M, $[H_2]$ = 0.10, $[H_2S]$ = 0.20M and $[CH_4] = 8.40 \times 10^{-5}$ M for the following reaction at 900°C, calculate the equilibrium constant (K_C)

$CS_2(g) + 4H_2(g) \rightleftharpoons CH_4(g) + 2H_2S(g)$

(A) 0.0120 (B) 0.0980

(C) 0.280 (D) 0.120

Q14. If 8 g mol of PCl_5 heated in a closed vessel of 10 L capacity and 25% of its dissociates into PCl_3 and Cl_2 at the equilibrium then value of Kp will be equal to:-

(A) P/30 (B) P/15

(C) 2/3P (D) 3/2P

Q15. For which reaction $\dfrac{K_P}{K_C}$ value is max ?

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(B) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(C) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(D) $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

Q16. For the reaction :

A ⇌ B, $K_C = 1$

B ⇌ C, $K_C = 3$

C ⇌ D, $K_C = 5$

K_C for the reaction A ⇌ D is :-

(A) 15 (B) 5 (C) 3 (D) 1

Q17. For the reaction

$A \rightleftharpoons B$; $K_1 = 2$

$B \rightleftharpoons C$; $K_2 = 4$

$C \rightleftharpoons D$; $K_3 = 6$

K_4 for the reaction $A \rightleftharpoons D$ will be :-

(A) 12 (B) $\frac{4}{5}$ (C) 24 (D) 48

Q18. $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

If $K_p = 64\ atm^2$ then equilibrium pressure of mixture is :-

(A) 8 atm (B) 16 atm

(C) 64 atm (D) 4 atm

Q19. The figure shows the change in concentration of species A and B as a function of time.

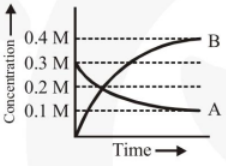
The equilibrium constant K_C for the reaction A(g) ⇌ 2B(g) is :-

(A) $K_C > 1$

(B) $K_C < 1$

(C) $K_C = 1$

(D) Data insufficient



Q20. $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

According to above reaction, partial pressure of CO_2 & CO at equilibrium are 4 & 8 respectively then find out K_p of the above reaction :-

(A) 6 (B) 2 (C) 16 (D) 32

Q21. Pure PCl_5 is introduced into an evacuated chamber and comes to equilibrium at 247°C and 2.0 atm. The equilibrium gaseous mixture contains 40% chlorine by volume

Calculate K_p at 247°C for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) 0.625 atm (B) 4 atm

(C) 1.6 atm (D) None of these

Q22. For the dissociation reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the degree of dissociation (α) in terms of K_p and total equilibrium pressure P is :-

(A) $\alpha = \sqrt{\frac{4P+K_p}{K_p}}$ (B) $\alpha = \sqrt{\frac{K_p}{4P+K_p}}$

(C) $\alpha = \sqrt{\frac{K_p}{4P}}$ (D) None of these

Q23. Which of the following is correct for the equilibrium of reaction-

$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

(A) $P_{(H_2)} \propto P_{(H_2O)}$ (B) $P_{(H_2)} \propto \sqrt{P_{(H_2O)}}$

(C) $P_{(H_2)} \propto P^2_{H_2O}$ (D) $P_{(H_2)} \propto \frac{P^2_{H_2O}}{P_{CO}}$

Q24. $A_{(g)} + 3B_{(g)} \rightleftharpoons 4C_{(g)}$ initial concentration of A is equal to that of B. The equilibrium concentration of A and C are equal. K_C of the reaction will be :-

(A) 0.8 (B) 8

(C) 80 (D) 0.08

Q25. The equilibrium constant in a reversible reaction at a given temperature :-

(A) Depends on the initial concentration of the reactants

(B) Depends on the concentration of the products at equilibrium

(C) Does not depend on the initial concentrations

(D) It is not characteristic of the reaction

Q26. Ammonium carbamate dissociates as $NH_2COONH_4(s)$ ⇌ $2NH_3(g) + CO_2(g)$

In a closed vessel containing ammonium carbamate in equilibrium, ammonia is added such that partial pressure of NH_3 now equals to the original total pressure. Calculate the ratio of partial pressure of CO_2 now to the original partial pressure of CO_2 :-

(A) 4 (B) 9 (C) $\frac{4}{9}$ (D) $\frac{2}{9}$

Q27. When 1 mole of pure ethyl alcohol (C_2H_5OH) is mixed with 1 mole of acetic acid at 25°C, the equilibrium mixture contains 2/3 mole each of ester and water :

$C_2H_5OH(l) + CH_3COOH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O$

The $\Delta G°$ for the reaction at 298 K is :

(A) 3435 J (B) 4J

(C) –3440 J (D) Zero

Q28. The equilibrium :

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ show that k_P (atm) is triple to the value of k_C(mol litre^{-1}) at a particular temperature T, then T is:-

(A) 24.36 K (B) 12.18 K

(C) 36.54 K (D) None

Q29. For the reaction (A) and (B)

$A(g) \rightleftharpoons B(g) + C(g)$ ...(A)

$X(g) \rightleftharpoons 2Y(g)$ ...(B)

Given $k_{P_1} : k_{P_2} = 6 : 1$

If the degree of dissociation of A(g) and X(g) be same then the total pressure at equilibrium (A) and (B) are in the ratio.

(A) 36 : 1 (B) 6 : 1

(C) 24 : 1 (D) 4 : 1

Q30. The equilibrium constants Kp_1 and Kp_2 for the reaction

$X(g) \rightleftharpoons 2Y(g)$ and $Z(g) \rightleftharpoons P(g) + Q(g)$, respectively are in the ratio of 1 : 4. If the degree of dissociation of X is 2 times that of Z, then the ratio of total pressure ($P_1 : P_2$) at these equilibria is (Assume degree of dissocation for both reaction are very very small) :

(A) 1 : 36 (B) 1 : 16

(C) 1 : 64 (D) None

Q31. The value of K_p for the reaction

$2H_2(g) + 2Cl_2O(g) \rightleftharpoons 4HCl(g) + O_2(g)$ is 0.03 atm at 427°C, when the partial pressure are expressed in atmosphere then the value of K_C for the same reaction is :

(A) 5.23×10^{-4} (B) 7.34×10^{-4}

(C) 3.2×10^{-3} (D) 5.43×10^{-5}

Q32. 2 moles of N_2 are mixed with 6 moles of H_2 in a closed vessel of one litre capacity. If 50% N_2 is converted into NH_3 till equilibrium, find the value of K_c for the reaction,

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(A) 4/27 (B) 27/4

(C) 1/27 (D) 27

Q33. Consider the equilibrium :-

$$CO_{2(g)} \rightleftharpoons CO_{(g)} + \frac{1}{2}O_{2(g)}$$

The equilibrium constant K is given by (when $\alpha << 1$)

(A) $K = \frac{\alpha^{3/2}}{\sqrt{2}}$ (B) $K = \frac{\alpha^3}{2}$

(C) $K = \frac{\alpha^{3/2}}{2}$ (D) $K = \frac{\alpha^{3/2}}{\sqrt{3}}$

Q34. The most stable oxide of nitrogen will be :-

(A) $2NO_2(g) \rightleftharpoons N_2(g)+2O_2(g)$; $K_{eq} = 6.7 \times 10^{16}$

(B) $2N_2O_5(g) \rightleftharpoons N_2(g) + 5O_2(g)$; $K_{eq} = 1.2 \times 10^{24}$

(C) $2NO(g) \rightleftharpoons N_2(g)+O_2(g)$; $K_{eq} = 2.2 \times 10^{30}$

(D) $2N_2O(g) \rightleftharpoons 2N_2(g)+O_2(g)$; $K_{eq} = 3.5 \times 10^{33}$

Q35. $A \rightleftharpoons 2B$ Kp_1, $C \rightleftharpoons D + E$ Kp_2

If degree of dissociation of A and C are same and $Kp_1 = 2Kp_2$, then the ratio of total pressure P_1/P_2 = ?

(A) 1/3 (B) 1/2 (C) 1/4 (D) 2

Q36. If $K_1 = 4 \times 10^{-3}$ for following gaseous reaction

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g); K_1$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g); K_2$$

Then K_2 will be :-

(A) 8×10^{-3} (B) 6.25×10^4

(C) 6.25×10^8 (D) 8×10^4

Q37. For the reaction : $2A(g) + B_2(g) \rightleftharpoons 2AB_2(g)$ The equilibrium constant, K_p at 300 K is 16.0. The value of K_p for $AB_2(g) \rightleftharpoons A(g) + \frac{1}{2}B_2(g)$ is :-

(A) 8 (B) 0.25

(C) 0.125 (D) 32

Q38. At 527°C, the reaction given below has K_C=4

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

What is the K_p for the reaction ?

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(A) $16 \times (800\ R)^2$ (B) $\left(\frac{800R}{4}\right)^{-2}$

(C) $\left(\frac{1}{4\times 800R}\right)^2$ (D) None of these

Q39. The equilibrium constant for the reaction :

$X_2 \rightleftharpoons 2X$ at 300 K and 500 K are 10^{-3} and 10^{-6} respectively. The reaction is-

(A) exothermic

(B) endothermic

(C) neither exothermic nor endothermic

(D) None

Q40. For a reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, the value of K_C does not depends upon :-

(a) Initial concentration of the reactants

(b) Pressure

(c) Temperature

(d) Catalyst

(A) Only c (B) a, b, c

(C) a, b, d (D) a, b, c, d

Q41. A sample of mixture of A(g), B(g) & C(g) under equilibrium has a mean molecular weight (observed) is 80.

The equilibrium is

$A(g) \rightleftharpoons B(g) + C(g)$

$(M_w = 100)$ $(M_w = 60)$ $(M_w = 40)$

Find the degree of dissociation (α) for A.

(A) 0.25 (B) 0.5

(C) 0.75 (D) 0.8

Q42. For which of the following reaction, product formation is favoured at low pressure and low temperature ?

(A) $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$; $\Delta H° = 172.5kJ$

(B) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH$; $\Delta H° = -21.7kJ$

(C) $2O_3(g) \rightleftharpoons 3O_2(g)$; $\Delta H° = -285\ kJ$

(D) $H_2(g) + F_2(g) \rightleftharpoons 2HF(g)$; $\Delta H° = -541kJ$

Q43. For the reaction $2A + B \rightleftharpoons C$, $\Delta H = +x$ Cals, which one of the following conditions would favour the yield of C on the basis of Le Chatelier principle?

(A) High pressure, high temperature

(B) Only low temperature

(C) High pressure, low temperature

(D) Only low pressure

Q44. Following two equilibrium is simultaneously established in a container

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

$$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

If some Ni(s) is introduced in the container forming $Ni(CO)_4(g)$ than at new equilibrium –

(A) $[PCl_3]$ will increase

(B) $[PCl_3]$ will decrease

(C) $[Cl_2]$ will remain same

(D) [CO] will remain same

Q45. For reaction $a\ A + b\ B \rightleftharpoons c\ C + d\ D$, if on increasing the temperature & pressure the degree of dissociation decreases then which is correct ?

(A) ΔH = +ve, $(a + b) > (c + d)$

(B) ΔH = –ve, $(a + b) > (c + d)$

(C) ΔH = +ve, $(a + b) < (c + d)$

(D) ΔH = –ve, $(a + b) < (c + d)$

Q46. In which reaction will an increase in the volume of the container favour the formation of products?

(A) $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

(B) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(C) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(l)$

(D) $3O_2(g) \rightleftharpoons 2O_3(g)$

Q47. The exothermic formation of ClF_3 is represented by the equation

$Cl_{2(g)} + 3F_{2(g)} \rightleftharpoons 2ClF_{3(g)}$; $\Delta H_r = -329$ kJ

Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2, F_2 and ClF_3?

(A) Removing Cl_2

(B) Increasing the temperature

(C) Adding inert gas at constant

(D) Decreasing the volume of the container

Q48. Which is in favour of forward reaction for reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ at constant temperature :-

(A) Addition of inert gas at constant volume

(B) Addition of $PCl_3(g)$ at constant volume

(C) Addition of $PCl_5(g)$ at constant volume

(D) Addition of $Cl_2(g)$ at constant volume

Q49. For which of the following reactions is product formation favoured by low pressure and high temperature ?

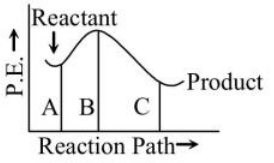
(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g); \Delta H^\circ = -9.4$ kJ

(B) $CO_2(g) + C(s) \rightleftharpoons 2CO(g); \Delta H^\circ = +172.5$ kJ

(C) $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH; \Delta H^\circ = -21.7$kJ

(D) $3O_2(g) \rightleftharpoons 2O_3(g)$; $\Delta H^\circ = 285$ kJ

Q50. With respect to the figure given below, which of the following statement is correct ?



(A) E_a for the forward reaction is C – B

(B) E_a for the forward reaction is B – A

(C) $E_{a(forward)} > E_{a(backward)}$

(D) E_a(for reverse reaction) = C – A

Q51. Ammonium carbamate (NH_4COONH_2) when heated to 200°C gives a mixture of NH_3 and CO_2 with a vapour density of 13.0 at equilibrium. What is the degree of dissociation of ammonium carbamate ?

(A) 3/2 (B) 1/2 (C) 2 (D) 1

Q52. $AB_3(g)$ is dissociates as

$AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2}B_2(g)$.

When the initial pressure of AB_3 is 800 torr and the total pressure developed at equilibrium is 900 torr. What percentage of $AB_3(g)$ is dissociated ?

(A) 10% (B) 20% (C) 25% (D) 30%

Q53. For the reaction $AB(g) \rightleftharpoons A(g) + B(g)$, AB is 33% dissociated at a total pressure of P. Then:-

(A) $P = K_P$ (B) $P = 4\ K_P$

(C) $P = 3K_P$ (D) $P = 8\ K_P$

Q54. Volume of the flask in which species are transferred is double of the earliest flask. In which of the following cases, degree of dissociation is affected -

(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(B) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(C) $PCl_5(g) \rightleftharpoons PCl_3 + Cl_2$

(D) (A) & (C) both

Q55. What is the effect of a catalyst in a reversible reaction :-

(A) Change the rate of backward reaction

(B) Changes the rate of forward reaction

(C) Changes the rates of forward and backward reactions equally

(D) None of the above

Q56. Standard Gibbs energy change ΔG° for a reaction is zero. The value of the equilibrium constant will be :-

(A) 1 (B) 10 (C) 100 (D) ∞

Q57. For the following reaction taking place at a certain temperature

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

if equilibrium pressure is X bar, then ΔG° will be :

(A) $-2RT\ln X$ (B) $-2RT\ln(X - \ln 2)$

(C) $-2RT(\ln X - \ln 2)$ (D) None

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. For gaseous homogenous reaction

$aA(g) \rightleftharpoons bB(g) + cC(g)$; $(b + c > a)$

K_p, K_c and K_x (eq. constant in terms of mole fraction)

Choose correct statements :-

(A) $K_p = K_c$

(B) K_p, K_c, K_x all depends only on temperature

(C) K_p and K_c depends on temperature while K_x depends on temperature and pressure both

(D) $K_p > K_c$

Q2. Choose the incorrect statements, for reversible reacton $R \rightleftharpoons P$:-

(A) kinetically K_f and K_b and E_f and E_b are comparable

(B) equilibrium can be attain in both direction

(C) decomposition of $CaCO_3(s)$ in closed vessel is a reversible reaction.

(D) at equilibrium only product is present

Q3. Von't hoff equation can be represent in the form of :-

(A) $\frac{d(\ell nK_p)}{dT} = \frac{-\Delta H^\circ}{RT^2}$

(B) $\frac{d(\ell nK_p)}{dT} = \frac{\Delta H^\circ}{RT^2}$

(C) $\ell nK_p = \frac{-\Delta H^\circ}{RT^2} + \text{constant}$

(D) $\ell nK_p = \frac{\Delta H^\circ}{RT^2} + \text{constant}$

Q4. For the reaction

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $\Delta H > 0$

The concentration of NO in the reaction vessel at equilibrium decreases when :-

(A) Temperature is increased

(B) Temperature is decreased

(C) Inert gas is introduced in the vessel

(D) A catalyst is introduced

Q5. The reaction(s) in which the yield of the product can be increased by the application of high pressure is/are :-

(A) $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

(B) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

(C) $N_2(g) + 3H(g) \rightleftharpoons 2NH_3(g)$

(D) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 6 to 8

The dissociation of nitrosyl chloride into NO and Cl_2 takes place according to the equation :

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

Varrying amounts of three gases were placed in a container and allowed to come to equilibrium at two different temperatures. The equilibrium concentrations of the three gases obtained are tabulated below :

Temperature	Concentrations is moles/lit		
	NOCl	NO	Cl_2
500 K	2.5×10^{-3}	1.5×10^{-3}	1×10^{-2}
800 K	4×10^{-4}	8×10^{-3}	2×10^{-4}

Q6. The expression for the equilibrium constant K, for the given reaction is -

(A) $K = \frac{[Cl_2][NO]^2}{[NOCl]^2}$

(B) $K = \frac{[Cl_2]^{1/2}[NO]}{[NOCl]}$

(C) $K = \frac{[Cl_2][NO]}{[NOCl]}$

(D) Both A and B are correct.

Q7. The ratio of equilibrium constants at 500 K to 800 K is –

(A) 4.5×10^{-2} (B) 45

(C) 45×10^{-2} (D) 450

Q8. With the introduction of inert gas to the given reaction at equilibrium at constant pressure –

(A) Forward reaction is favoured

(B) Backward reaction is favoured

(C) There is no effect on equilibrium

(D) None of these

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q9. Match list I with list II and select the correct answer:

List–I	**List–II**
(A) Reaction is reversed	(P) $\sqrt{K}$
(B) Reaction is divided by 2	(Q) K^2
(C) Reaction is multiplied by 2 and reversed	(R) 1\K
(D) Reaction takes place in two steps, each with same equilibrium constant	(S) $1/K^2$

(A) A→R ; B→S ; C→P ; D→Q (B) A→R ; B→Q ; C→S ; D→P

(C) A→R ; B→P ; C→S ; D→Q (D) A→P ; B→Q ; C→S ; D→R

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q10.

Column I	**Column II**	**Column III**
(A) Pressure increased in $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$ Eq. Pr. 2.5 atm 1atm 2.5 atm	(P) Forward shift of equilibrium	(i) $K_P = 2$
(B) Pressure increased in $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$ Eq. Pr. 1 atm 1atm 2 atm 1 atm	(Q) Backward shift of equilibrium	(ii) $K_P = 0.4$
(C) Temperature and pressure both increased in $3O_2(g) \rightleftharpoons 2O_3(g)$ Eq. Pr. 2.5 atm 2atm ; $\Delta H = 285$ kJ	(R) Equilibrium is unaffected	(iii) $K_P = 1$
(D) Pressure decrease and moles of N_2 increased $N_2(g) + 2O_2(g) \rightleftharpoons 2NO_2(g)$ Eq. Pr. 1 atm 2.5 atm 2.5 atm ; $\Delta H = 66.4$ kJ	(S) Theoretically we can not predict	(iv) $K_P = 0.25$

(A) B, (i), (P) (B) C, (iv), (S) (C) A, (ii), (R) (D) C, (iv), (Q)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. The equilibrium constant K_c for the reaction $P_4(g) \rightleftharpoons 2P_2(g)$ is 1.2 at 300°C. Suppose that 4 moles of $P_4(g)$ and 2 moles of $P_2(g)$ are mixed in 2 litre container at 300°C.

Find the value of reaction quotient (Q) and multiply it with 2.

Q12. Calculation the equilibrium constant for the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$.

If in equilibrium state at 3800K 50% nitrogen is converted into nitric oxide.

Q13. Two solid compounds A_1 and A_2 in a container dissociates at temperature T to show the equilibrium:-

$A_1(s) \rightleftharpoons A(g) + 2B(g) \quad K_{P_1} = 9\times10^{-3}$

$A_2(s) \rightleftharpoons C(g) + 2B(g) \quad K_{P_2} = 4.5\times10^{-3}$

The total pressure of gases at equilibrium will be.

Note : If your answer is 0.25, write in the form of addition as 0 + 2 + 5 = 7

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
A	B	D	A	B	B	B	D	B	B
11	12	13	14	15	16	17	18	19	20
B	A	B	D	B	D	B	D	D	D
21	22	23	24	25	26	27	28	29	30
C	D	C	B	A	D	A	A	D	B
31	32	33	34	35	36	37	38	39	40
B	A	D	B	A	A	A	C	B	A
41	42	43	44	45	46	47	48	49	50
B	B	B	B	C	C	B	B	B	C
51	52	53	54	55	56	57			
A	C	A	D	A	B	B			

Ex-1A

1. A, B **2.** A **3.** B, D **4.** A, B, C **5.** C
6. A, B, C, D **7.** B, C **8.** B, D **9.** B **10.** A
11. C **12.** D **13.** (A) → Q ; (B) → R ; (C) → P, S; (D) → Q
14. 8 **15.** 3

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	A	B	C	C	D	D	B	C	D
11	12	13	14	15	16	17	18	19	20
D	A	C	B	C	A	D	B	A	C
21	22	23	24	25	26	27	28	29	30
C	B	B	B	C	C	C	C	C	C
31	32	33	34	35	36	37	38	39	40
A	A	A	A	B	B	B	C	A	C
41	42	43	44	45	46	47	48	49	50
A	C	A	B	D	A	D	C	B	B
51	52	53	54	55	56	57			
D	C	D	D	C	A	C			

Ex-2A

1. C, D **2.** D **3.** B, C **4.** B **5.** A, C, D
6. A **7.** A **8.** A **9.** C **10.** C
11. 1 **12.** 4 **13.** 9

-: Important Notes & Formulas :-